2019 IEEE 20th International Conference on Information Reuse and Integration for Data Science (IRI)

Context-based Knowledge Discovery and Querying for Social Media Data

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Abstract

Modern Early Warning Systems (EWS) rely on scientific methods to analyse a variety of Earth Observation (EO) and ancillary data provided by multiple and heterogeneous data sources for the prediction and monitoring of hazard events. Furthermore, through social media, the general public can also contribute to the monitoring by reporting warning signs related to hazardous events. However, the warning signs reported by people require additional processing to verify the possibility of the occurrence of hazards. Such processing requires potential data sources to be discovered and accessed. However, the complexity and high variety of these data sources makes this particularly challenging. Moreover, sophisticated domain knowledge of natural hazards and risk management are also required to enable dynamic and timely decision making about serious hazards. In this paper we propose a data integration and analytics system which allows social media users to contribute to hazard monitoring and supports decision making for its prediction. We prototype the system using landslides as an example hazard. Essentially, the system consists of background knowledge about landslides as well as information about data sources to facilitate the process of data integration and analysis. The system also consists of an interactive agent that allows social media users to report their observations. Using the knowledge modelled within the system, the agent can raise an alert about a potential occurrence of landslides and perform new processes using the data sources suggested by the knowledge base to verify the event.

Keywords— early warning system; landslide hazard; high variety data; IoT; ontology; data sources discovery

1 Introduction

The application of Early warning Systems (EWS) to predict natural hazards and vulnerabilities plays a vital role in preventing loss of life and damage to property. For effective and timely decision-making, EWS requires a strong technical underpinning and sophisticated knowledge of the natural hazards and risk management. Landslides are a commonly occurring natural hazard with global impact and is closely linked to many other natural hazards such as earthquakes, storms, flooding and volcanic eruptions. Predicting individual occurrences of landslide events is complex as it depends upon many local factors, variables and anthropogenic input. For predicting and monitoring landslides, decision makers use scientific models to analyse Earth Observation (EO) data from satellites and ancillary data produced by Internet-of-Things (IoT) sensors deployed in landslide-prone areas. In addition, the ancillary data includes sensor data, social media data and data from other sources which are essential for the prediction and monitoring of hazard events. Such EO and sensor data used for the analysis are usually obtained from multiple and heterogeneous data sources. Furthermore, through social media channels (Facebook, Twitter, Instagram etc.) the general public can also contribute to landslide monitoring by reporting observations that could be warning signs for landslides. However, decision-makers need to verify the detected events reported from social media by analysing sensor or other corroborating data from the area of interest. Hence, EO and data representing the event in the area of interest is essential for the verification.

The high volume and variety of data produced by multi-

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ple and heterogeneous data sources (such as sources of social media and IoT sensors data) means that the most relevant sources need to be identified for analysis. A common knowledge base that captures and formally represents the core concepts (such as hazards, warning signs, data sources) and the relationships between them could be utilised to discover the potential data sources for analysis is therefore a critical component of EWS with complex data processing requirements. Moreover, cross-domain analysis of data is not only useful for the discovery of event co-relation but also allows for additional event verification. The process of landslide prediction and monitoring in EWS leads to challenges in (i) *knowledge creation and discovery* — as a formal knowledge base due to the large volume of user-generated micro texts from social media and the exploration of meaningful information from the knowledge base to support the decision making; (ii) *ad-hoc data analysis* — due to the verification process being dependent on the context and completeness of information provided by the users; and (iii) *data sources discovery* — due to the heterogeneity of sensor types and a wide spectrum of performance and accuracy characteristics in sensor data streams.

The main *contribution* of this work is a formal knowledge base of landslide domain concepts to enable the integration of EO and ancillary data from multiple heterogeneous sources for dynamic verification and early prediction of landslide events based on context and information reported by social media users. The underpinning of this knowledge base is the Landslip ontology that captures the correlations between landslides, multi-hazards, warning signs, sensor data and other data sources. The purpose of the ontology is to facilitate data discovery, which will be used to find potential data sources for landslide verification. Another contribution of this paper is Onto-DIAS, a prototype system that utilises the knowledge base to support data integration and analytics in landslide early warning applications.

The rest of this paper is organised as follows: related work is discussed next in section 2, followed by the details of Landslip ontology in section 3. Populating the knowledge base from social media content and the ontology is described in section 4. The system architecture and implementation are described in sections 5 and 6 respectively. Finally, the paper is concluded in section 7.

2 Related Work

In this section, we present related work on data integration and analytics, semantic web technologies, and natural language processing in multi-hazard EWS.

2.1 Data Integration and Analytics In multi-hazard Early Warning Systems

EWS have played a significant role in natural hazard management to minimise loss of life and damage to property. Additionally, several modern multi-hazard EWS take advantage of various type of data sources, including remote sensing satellites, IoT sensors, and social media. Due to the heterogeneity of the data sources, data integration becomes a vital part of EWS to provide high-quality data for the effective prediction of hazard events. Several works on data integration and analysis for multi-hazard have been proposed in the literature. A data analytic framework for Twitter data was proposed in [35] to identify twitter messages that are related to a particular type of disaster (e.g. earthquake, flood, and wildfire). Several methods, including matching-based and learning-based, to identify relevant tweets, are also evaluated. In [16], the authors describe a study on the identification of peatland fires and haze events in Sumatra Island, Indonesia, by using social media data. A data classification algorithm is applied to analyse the tweets, and the outcomes are verified by using hot spot and air quality data from NASA satellite imagery. The authors in [33] propose an information infrastructure for timely delivery of social media and crowd-sourcing message (from Ushahidi platform) to potentially responsible departments during the Haiti earthquake in 2010. A data classification algorithm is used to provide an automatic classification mechanism over the messages. A decision support system that integrates crowd-sourcing data with Wireless Sensor Networks (WSN) to widen the coverage of the monitoring area for flood risk management in Brazil is proposed in [14]. The Open Geospatial Consortium (OGC) standards are used in the research to aid in the integration of the crowd-sourced data.

2.2 Semantic web technologies for multi-hazards management

Semantic Web technologies have been widely used in hazard management to present knowledge of hazards and their relation to EO and ancillary data. The application of ontology for hazard management includes hazard assessment and urbanisation analysis. Here, the Semantic Sensor Network Ontology (SSN) [30] and the Semantic Web for Earth and Environmental Terminology (SWEET) [32] are two significant ontologies that can be applied for hazard management. In [32], the authors reuse SWEET to conceptualise knowledge and expertise in several areas, such as buried assets (e.g. pipes and cables), roads, soil, the natural environment and human activities. Additionally, they propose to use the Ontology of Soil Properties and Process (OSP) to describe soil properties such as soil strength and

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processes such as soil compaction. The OSP and other concepts are used to express how they affect each other in asset maintenance activities. The SSN ontology is used in [30] and [7] for wind monitoring. The former uses SSN with Ontology for Kinds and Units (QU) [21] to conceptualise wind properties (e.g. wind speed and direction) whereas the latter uses SSN and SWEET to model the concepts of wind sensor and data streams of wind observation. In [25], the authors extend the SSN ontology to create an ontology for landslides including relevant concepts such as earthquake, geographical unit, soil, precipitation, and wind. Even though these ontologies provide general concepts for sensor data and hazard event, they cannot describe concepts of human sensors (e.g. social media data). Further processes are needed when applying these ontologies to EWS for multi-hazard applications.

2.3 Natural Language Processing

Natural Language Processing (NLP) is a set of information engineering techniques which enables computers to process and make sense of human (natural) languages. NLP technique has evolved from complex handwritten rules to models trained using machine learning. Earlier machine learning techniques like decision trees[4, 23] generated rules similar to handwritten ones, using machine learning. The application of NLP in this work is to extract useful information from the natural language and to classify the content into different topics of interest. Language modelling techniques apply probability distribution over a sequence of words. Unigram, n-gram [6, 8], Exponential and Neural networks [26, 5, 18] are the main types of language models in use. Recent studies promise high accuracy in classifying natural language using a neural network. A unified architecture for NLP using deep learning technique has been introduced in work [9] by NEC Labs. In this work, the input sentence can be processed to perform part-of-speech tagging, chunking, named entity tags, semantic roles etc. using a language model and CNN. A study [17] at New York University reveals a series of experiments using Convolutional Neural Network (CNN) which is trained on a pre-trained model of word vectors for sentence classification in which the model showed significant improvement in performance in several NLP tasks. Over the years several open-source NLP projects like NLTK[22], CoreNLP[24], Spacy[1], GATE[11] etc. gained interest of both academia and industry. While these methods and tools support natural language processing, building knowledge from natural language pose several challenges.

3 Landslip Ontology

Earth Observation (EO) and ancillary data provided by multiple data sources contribute to the high variety of characteristics of data sources in EWS. In addition, such data sources differ in terms of: (i) data type — the different types of EO and ancillary data used in a particular analysis, e.g. soil properties, temperature, humidity; (ii) data storage — the difference in methods for collecting and organising data, e.g. RDBMS, NoSQL database, and distributed file system; and (iii) data access — data sources are accessible by different methods, ranging from direct access through data stores (e.g. JDBC) to standard Web Services (e.g. OGC, SOAP, Restful). These differences make the discovery, access and integration of data in EWS quite challenging. A formal semantic representation of the data sources, domain knowledge about natural hazards and the relationship between them can help address the challenges arising from such differences and enable data integration and analysis.

The knowledge base for data integration and analysis is designed and developed based on the principles of Semantic Web Technology [38]. A core component of the knowledge base is an ontology, which can be can be defined as ”a formal, explicit specification of a shared conceptualisation” [34]. That is, an ontology models the agreed knowledge about the real world through explicitly defined concepts and constraints on them that are machine readable. In this work, we have developed an ontology, namely the Landslip Ontology, which is based on OWL 2 [37]. The Landslip Ontology contains knowledge on the relationships between landslides and data sources for EO and other data. Additionally, it also contains the domain knowledge for describing the interaction of landslide event to other hazards and the warning signs, which can be precursors to landslides. Figure 1 shows the steps of the Landslip ontology development. The primary knowledge sources for designing the ontology came from interviews with four scientists and experts in landslide hazard management with an average of 10 years of experience between them. Besides, publications [12], [13] and standard specifications [30, 32, 27, 28, 29] involving multi-hazards and geospatial data models were also used as knowledge sources to design the ontology. A scenario-based approach [19] was used to define a narrative representing expected uses of an EWS in the domain of interest from the viewpoint of both domain experts as well as ontology developers. The scenario helps to define the scope of the domain ontology to be designed and frame competency questions to model the domain knowledge as well as for evaluating the ontology [3, 36]. The Landslip ontology is developed and imported to a triple store as a model for building the knowledge base for Landslide multi-hazard EWS. Finally, a knowledge API is developed, which provides the consumer with an access point to the knowledge

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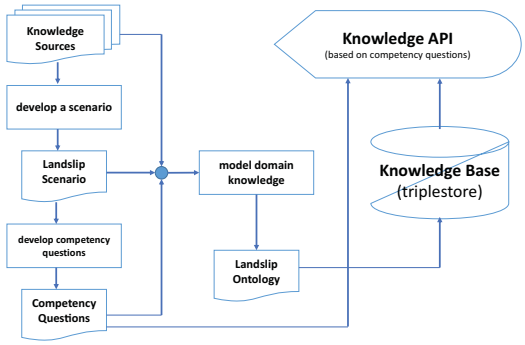

Figure 1. Landslip Ontology development process

base.

Figure 2 depicts a section of the Landslip Ontology consisting of domain concepts and the relationships between the concepts. The ontology is comprised of two modules: (i) *Landslide Hazard Ontology* – defines concepts about landslides and the interaction to other hazards and corresponding warning signs; and (ii) *Data Sources Ontology* – defines concepts about observation and data sources for landslide hazard risk assessment. The Landslip ontology reuses the SSN ontology as well as terminologies defined in OGC standards (e.g. Observation and Measurement [27], SensorML [28] and SOS [29]).

The ontology was evaluated for consistency and correctness through competency questions and a set of synthesised data that represent the use case(s) of landslide hazard. The competency questions were defined in SPARQL [39] and the knowledge base queried for answers. An example competency questions is *"What are data sources and their metadata to observe a set of hazards H1, H2, Hn"* and the corresponding SPARQL query to answer this question is defined as follows:

```
SELECT  ?hazard ?observation
    ?observedProperty
    ?dataSource ?metadata
    ?profile ?predicate ?_value
WHERE {
    ?observation :isObservationFor
    ?hazard .
    ?observedProperty
    :isObservedPropertyFor ?observation .
    ?dataSource :isDataSourceFor
    ?observedProperty .
    ?dataSource :hasDataSourceMetadata
    ?metadata . ?metadata :hasProfile
    ?profile . ?profile ?predicate ?_value .
    VALUES (?hazard)
        { (:flood_1) (:landslide_1) } .
    FILTER (?p != rdf:type)
}
```

4 Populating the Knowledge Base from Social Media Content

The Landslip Ontology is a conceptual model that formally represents domain knowledge about landslides captured from domain experts of natural hazards management. Th ontology consists of concepts and relationships but does not model concrete objects or *named individuals*, that represent actual events of landslides. With the emergence of social media as a potential resource to build the domain knowledge, social media contents to represent actual events of landslides are dynamically instantiated within the ontology. In order to do this however, sophisticated techniques are required to understand the context of the social media content and extract information from the content to create individuals based on the conceptual model. Figure 3 shows the process of populating the knowledge base from social media content to facilitate landslide-related knowledge discovery in an EWS. Historical data related to past events of landslides collected from social media platforms are also added to the knowledge base. The knowledge base thus consists of a set of synthesised data for hazards (landslides, flood), warning signs (e.g. leaning light pole, blocked road), and EO and ancillary data (e.g. water potential, moisture and temperature). Due to the wide variety of information provided in each text from the collected social media data, classification of the text into the topics of interest and extraction of useful information from the content are required. One of the most critical challenges while dealing with user-generated content is to capture the semantics of the content using Natural Language Processing (NLP)/ Natural Language Understanding (NLU) techniques. Our prototype system is designed with two modules to achieve this task: A data classification/topic detection module for social media content classification and a data extraction module to extract useful information, which can be used for instantiating objects in the ontology. The modules are described next.

4.1 Social Media Content Classification

The first step to process the user text is to identify the hazard which the user is referring to on social media. Recurrent Neural Network (RNN) [20] such as Long Short Term Memory networks (LSTM) and Convolutional Neural Network (CNN) are widely used in text classification. In this work, we have used CNN, a deep learning technique, [17] to perform text classification. A model, which is custom

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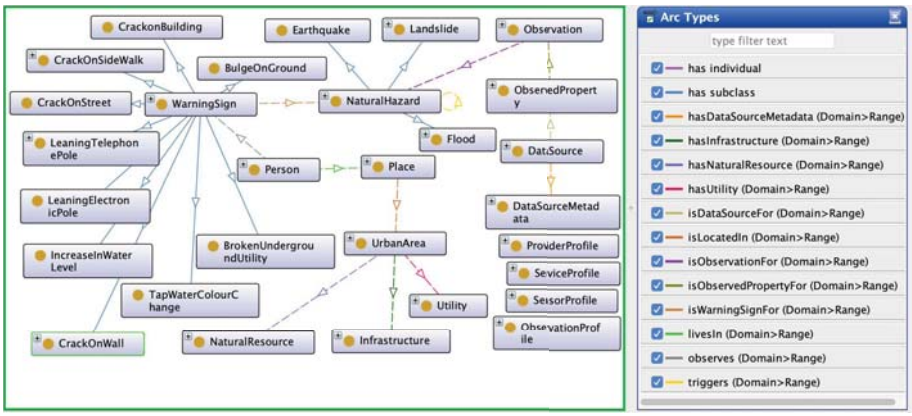

Figure 2. Section of Landslip Ontology.

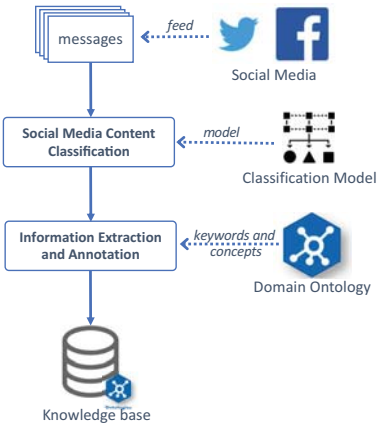

Figure 3. Process of populating the Knowledge Base from social media content

trained using weather related text is used. The model training and inference processes are explained in the following sections.

4.1.1 Model

A classification hierarchy, as shown in Figure 4, has been defined for our prototype system. A model is trained using 1000 user-generated texts related to hazards, which are marked for different hazard events and warning signs (for example, Events: Flood, Heavy Rainfall, Snow etc.; Warning Signs: Leaning Light Pole, Water Discolouration etc.). Each class has around 200 records. TensorFlow [2], an open source library, was used for data preparation, training and inference. The model is similar to the one proposed by Kim Yoon in his work Convolutional Neural Networks for Sentence Classification [17], which achieved good classification performance for different text classification tasks like sentiment analysis and is a standard baseline for new text classification methods. The model consists of a word embedding layer, which maps vocabulary word indices to lower dimensional vector spaces. The convolutional layer calculates convolutions over the embedded word vectors using different filter sizes as each convolution produces tensors of different shapes followed by max-pooling, which is a sampling-based discretization process. These vectors are later merged to form a large feature vector. Full details of the CNN layers and training process are beyond the scope of this paper.

4.1.2 Inference

Every text message received by the Landslip agent is passed to the classifier, which outputs the hazard event and any warning sign mentioned in the text. This step enables the system to understand the topic from the user-generated content. Data classification for this system is a two-step process involving the classification of hazards and warning signs. In the first step, the classifier tags the message whether a hazard or warning sign is present in the text. The second step involves two classifiers, one for classifying the type of hazard and the second for categorising the kind of warning sign as per the classification hierarchy. In some cases, a message may contain information about both hazard and warning sign. In such a scenario, the system passes this message to both classifiers. In the inference step, the result is attached as metadata to the input text.

4.2 Information Extraction and Annotation

In this step, a scenario is developed using information about the situation from user-provided text. NLP techniques, namely Part of Speech (PoS) tagging and Named Entity Recognition (NER) are used to extract useful information from the text. Pre-trained NLP models for the

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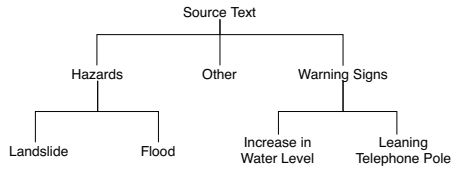

Figure 4. Data Classification Hierarchy

English language recognise Geo-location and affected entities (for example, Road, Building, Electric Pole, etc.). The English language model,a multi-task CNN trained on OntoNotes[15], with GloVe[31] vectors trained on Common Crawl[10] is used in this work. This model is built for assigning word vectors, context-specific token vectors, POS tags, dependency parse and named entity recognition.

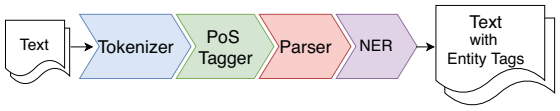

Figure 5. NLP Pipeline for Named Entity Recognition

As mentioned in Section 2, named entities are extracted from the user-generated content using NER. An NLP tool called Spacy [1] was used to perform the series of tasks required to perform NER. The processing pipeline consists of a tokenizer, PoS tagger, Parser, and NER. The tokenizer tokenizes sentences into words for which a PoS tag is attached based on the sentence structure. Then the parser performs a dependency parsing of the sentence, which represents its grammatical structure and defines the relationship between words. This step is followed by the NER phase, which identifies the type of entity such as geo-location, person, organisation, physical object, date, time, building/infrastructure etc. This model and pipeline gave a largely accurate prediction of the type of the entity from the noun words tagged from the PoS tagging phase. Figure 6 shows an example of a sentence being processed and labelled. The entity tags are

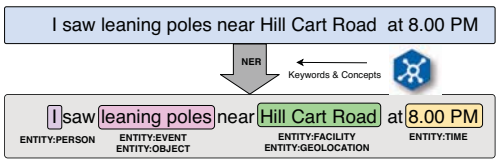

Figure 6. Named Entity Recognition Example

attached to the original data as metadata for storage and indexing. This extracted information is instantiated as objects based on the concepts defined in the ontology and stored in the knowledge base.

5 Onto-DIAS Architecture

The main aim of the prototype, Onto-DIAS, is to provide a comprehensive suite of services to enable multi-hazard early warning through timely detection of warning signs. Incidents reported by social media users are used as an initial input to detect warning signs for natural hazards. However, additional processes are required to verify or predict the potential of hazards. Data sources discovery and data access are essential functions for searching, acquiring and integrating EO and ancillary data from potential data sources to be used in the processing. Onto-DIAS system utilises Landslip ontology and its knowledge base to facilitate data integration over multiple data sources.

Figure 7 (a) presents the architecture of Onto-DIAS, which comprises of the following components:

- *Data sources* — Data sources collect EO and ancillary data from physical sensors deployed in the landslide-prone area. These sensors observe or measure properties of landslide and other earth observations, which can be used to warn about the potential occurrence of landslides.

- *Knowledge APIs* — Knowledge APIs are RESTful Web Services that provide access to the landslide knowledge base to support data integration and analysis. The APIs comprise of 3 main APIs: (i) *hazard knowledge API* — to answer competency questions which are defined for the Landslip Ontology. This API also provides a function for hazard and warning sign indicators from social media input. It is implemented using CNN to create a data classification model with social media data used for the training phase; (ii) *data sources discovery API* — provides a function for searching potential data sources; and (iii) *data access API* — This API utilises information given by the data sources discovery API to access and query data from multiple data sources and provide the data for the data analytics process.

- *Landslip agent* — The Landslip agent communicates with users via a social media platform to receive information about landslide-related incidences from them. An agent for the Facebook platform was developed for Onto-DIAS. The agent uses the knowledge API to understand text messages from users. It then decides whether to ask users for additional relevant information or execute analytic processes to verify or predict the occurrence of a landslide.

- *User application* — A User application allows social media users to report hazard-related incidences and additional information such as observed date, location etc. For Onto-DIAS, Facebook Messenger was used

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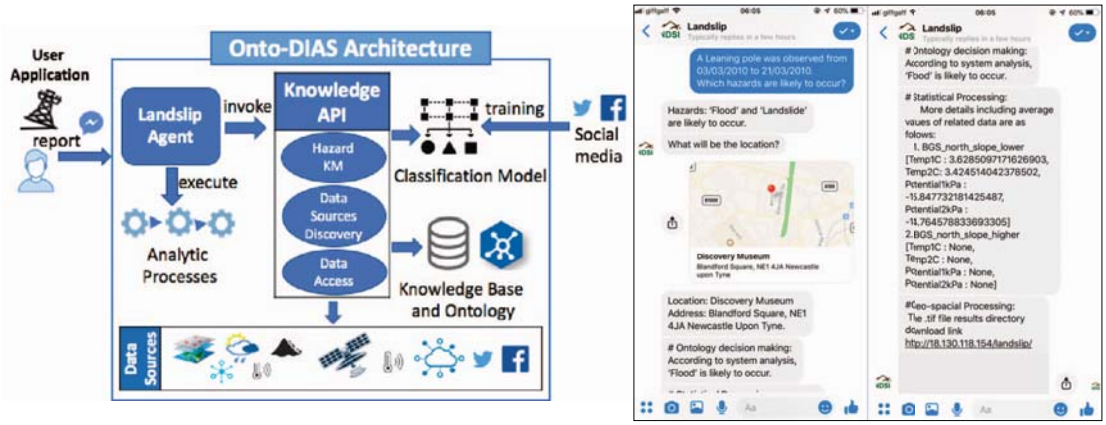

(a) (b)

Figure 7. (a) Onto-DIAS Architecture; (b) A communication between social media user and Onto-DIAS

as the user application to enable social media users to communicate with the Landslip agent.

6 Implementation

Onto-DIAS system is developed and deployed in a cloud server. Some samples of EO and urban data are stored in different data stores (e.g., MySQL, MongoDB, and HDFS) to demonstrate the heterogeneity of data sources and to conduct experiments. Communication between social media users and Onto-DIAS is demonstrated using a mobile version of the Facebook Messenger application. A Facebook group, *Landslip*, was created for the experiment. Social media users can search and join the group to contribute by reporting warning signs related to landslides. When the agent receives a message from an individual user via the Facebook group, it starts communication to acquire more information and perform data analysis to conduct landslide events verification. Figures 7 (b) depicts examples of communication between a social media user and Onto-DIAS. A user sends a text message via the Landslip Facebook group to report an observed incidence. On receiving the message, Onto-DIAS agent accesses the knowledge base through the knowledge APIs to indicate related hazard events and respond to the user. If the report from the user involves a hazard, the agent will ask for additional information related to the incidence (e.g. observed date and location). Once the agent obtains enough information, the knowledge base is utilised to search for potential data sources related to the reported incidence. The list of data sources suggested from the knowledge base is returned and accessed by Onto-DIAS to perform data analysis for hazard event verification. As a result, the agent returns the data analysis output to the user along with further information depending on the analysis models.

7 Conclusion and Future Work

A comprehensive set of EO and ancillary data from multiple and heterogeneous data sources is essential for an effective early warning system (EWS). In this paper, we demonstrated the application of ontology-based data integration and analysis for landslide reporting and detection through the development of a prototype Early Warning System (EWS), Onto-DIAS. Onto-DIAS provides a knowledge base for the landslide domain, which can be utilised for discovering potential data sources to verify the possibility of the occurrence of landslide based on the information reported by social media users. Using a social media platform, an agent interactively communicates with social media users who want to report any landslide warning sign they observed. This allows the EWS to make dynamic and timely decision regarding the possibility of the occurrence of landslide. Our immediate future work on this research is to evaluate Onto-DIAS by implementing in a real application of landslide EWS.

8 Acknowledgments

This research is partially supported by two Natural Environment Research Council projects including LandSlip (NE/P000681/1) and FloodPrep (NE/P017134/1).

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